them in circular fashion. and the 2 company people themselves can be arranged in $2!$ ways.

$\Rightarrow (15-2+1) = 14$ groups

Circular arrangements $= (14-1)! \cdot 2! = 13! \cdot 2!$

(14) How many words can be formed with "vector" such that (i) all vowels together?

Ans:- vector $\rightarrow$ 2 vowels (e & o) together as a group & in themselves $2!$ ways

$\Rightarrow (6-2+1)! \cdot 2! = 5! \cdot 2!$

(ii) all vowels @ odd places?

$\underline{\quad}_{1^{st}} \; \underline{\quad}_{3^{rd}} \; \underline{\quad}_{5^{th}} \; \underline{\quad}$

total 2 vowels (e & o) can be placed in odd places nothing but 3 places we have. here $r > n$ $(r=3, n=2)$ then we have to swap them consider as ${}^3P_2$ ways

$\therefore$ 2 letters can be placed in 2 places of 3 in ${}^3P_2$ ways $= 3 \times 2 = 6$ ways.

Remaining 4 letters in 4 places irrespective of even (or) odd places in. ${}^4P_4$ ways $= 4! = 24$ ways

$\therefore$

$\underline{\quad}_{1^{st}} \; \underline{\quad} \; \underline{\quad}_{3^{rd}} \; \underline{\quad} \; \underline{\quad}_{5^{th}} \; \underline{\quad}$

6 ways $\qquad 4! = {}^4P_4 = 24$ ways

$= 6 \times 24 = 144$ words we can get.

(14) A person has 12 friends out of which 7 are relatives. In how many ways can he invite 6 friends such that at least 4 of them are relatives?

Ans:- No. of non-relatives = 12 − 7 = 5

He may invite 6 friends in the following ways.

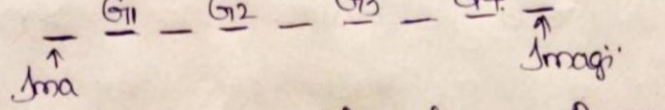
(i) 4 relatives + 2 non-relatives = $^7C_4 \times {}^5C_2$ (or)

(ii) 5 relatives + 1 non-relative = $^7C_5 \times {}^5C_1$ (or)

(iii) 6 relatives + 0 non-relative = $^7C_6$ ⇒

∴ total ways = $^7C_4 \times {}^5C_2 + {}^7C_5 \times {}^5C_1 + {}^7C_6$

(15) Find the no. of ways in which four girls & 3 boys can arrange themselves in a row so that none of the boys is together? How is this arrangement different from that in a circular way?



(i) Linear Arrangement / ~~Seat~~ linear Permutation:-

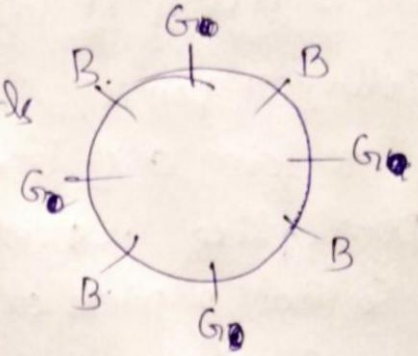
Above diagram represents the arrangement according to the condition.

first 4 girls in ($G_1 G_2 G_3 \& G_4$) 4 places in $^4P_4 = 4!$ ways

Boys in such a way that no boy together means 5 places are possible for 3 boys (including 2 assumption places) in $^5P_3$ ways (Actually $n=3, r=5$ but we have to swap when this kind of scenario comes).

∴ Total ways = $^4P_4 \cdot {}^5P_3 = 24 \times 60 = 1440$ ways.

(ii) Circular arrangement:-



In a circular arrangement 4 girls @ 4 places in $(4-1)!$ ways.

Here none of the boys / no two boys together means, the boys should be in b/w girls only. So, that 4 places & 3 boys to be arranged (swap n & r values because $n<r$ & to make $n>r$) in ${}^4P_3$ ways $= 4!$ ways.

Totally $= 4! \cdot {}^4P_3 = 4! \cdot 4! = 24 \times 24 = 576$ ways < 1440 ways

always circular permutations < linear permutations

(16) How many shakehands in the party of 10 members in such a way that every person shakes hand with every other person?

Ans:- shake hand nothing but '2' hands nothing 2 people to be selected for 1 shakehand like wise

Total shake hands $= {}^{10}C_2 = \frac{10!}{(10-2)! \cdot 2!} = \frac{10 \times 9 \times 8!}{8! \cdot 2} = 45$

45 shakehands.

(17) In a party if total 105 shakehands happened b/w every person with every other person, then total no. of members in the party?

${}^nC_2 = 105 \Rightarrow \frac{n(n-1)(n-2)!}{(n-2)!\,2!} = 105 \Rightarrow \frac{n(n-1)}{2} = 105 \Rightarrow n(n-1) = 210$

$n(n-1) = 210 \Rightarrow n^2 - n - 210 = 0 \Rightarrow n^2 - 15n + 14n - 210 = 0$

$\Rightarrow n(n+15) + 14(n-15) = 0 \Rightarrow (n-15)(n+14) = 0 \Rightarrow \boxed{n = 15}$